



2008

Mathematics

Trial Examination

HSC Assessment Task 4

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question
- START EACH QUESTION ON A NEW SHEET OF PAPER AND WRITE YOUR NAME ON THE SHEET.
- A table of Standard Integral has been provided at the back of the examination booklet.
- Students may use a curve drawing template which does not contain printed formulae other than equations of simple curves that may be drawn using the template.

Total Marks - 120

- Attempt all questions from 1 – 10
- All questions are of equal value

Total marks – 120 marks**Attempt Questions 1 – 10****All questions are of equal value**

Answer the questions on your own paper or writing booklet, if provided. Start each question on a new page.

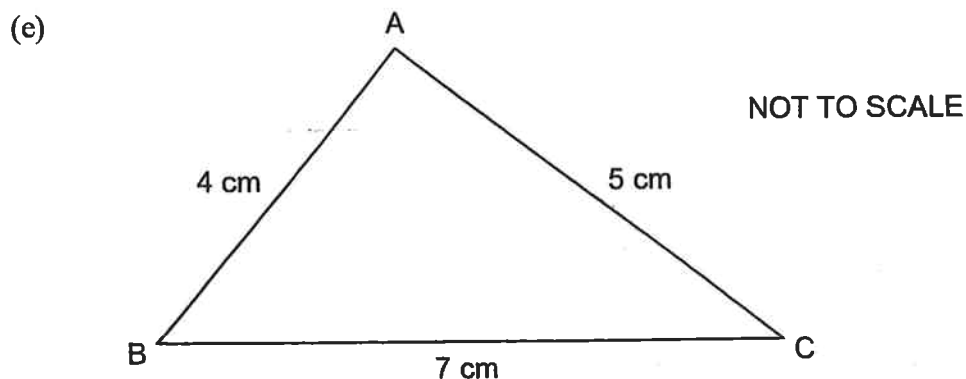
Question 1 (12 marks) Use a SEPARATE page or writing booklet**Marks**

(a) Evaluate: $\sqrt[3]{e^{2.4} - 1}$ correct to 5 significant figures. 2

(b) Given: $\frac{5}{\sqrt{3} - 1} = a\sqrt{3} + b$, find the values of a and b . 2

(c) Solve, giving your answer(s) in exact form: $2x^2 - 5x - 4 = 0$. 2

(d) Find a primitive of: $\frac{1}{x^2} + \frac{1}{x}$. 2



In the diagram above, find the size of the largest angle.
Give your answer correct to the nearest degree.

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(f) Simplify: $\frac{5}{m-2} - \frac{2}{m-3}$. 2

Question 2 (12 marks) Use a SEPARATE page or writing booklet**Marks**(a) Differentiate with respect to x :

(i) $\frac{\cos x}{x-1}$.

2

(ii) $(3x^2 - 7)^5$.

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(b) Solve: $|2x - 3| < 1$.

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(c) (i) Find: $\int \frac{x}{x^2 + 2} dx$.

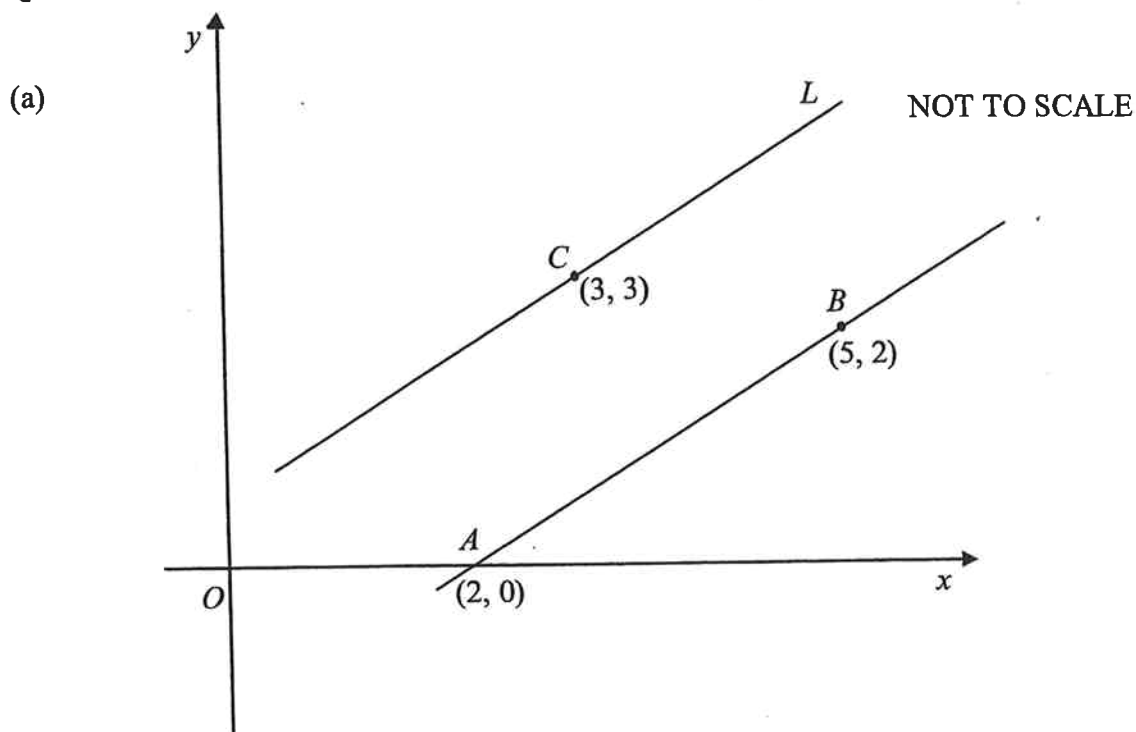
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(d) Use the change of base rule to evaluate: $\log_8 4$.

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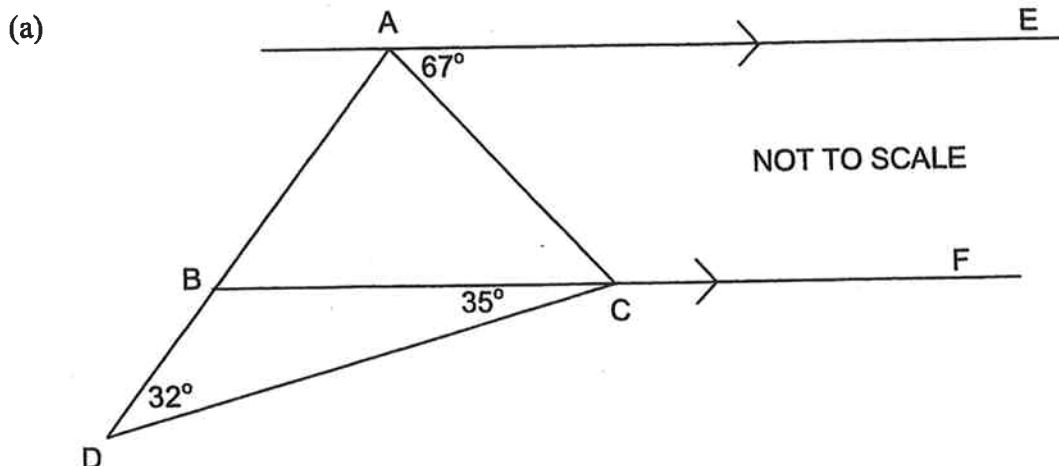
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| (i) Find the exact length of AB. | 1 |
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| (v) D is a point on L such that the length of DC is $\frac{\sqrt{13}}{2}$ units. What type of quadrilateral is ABCD? Give reasons. | 1 |
| (vi) Calculate the area of ABCD. | 1 |
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| (b) Solve: $2 \cos A = -\sqrt{3}$, for $0 \leq A \leq 2\pi$. | 2 |
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In the diagram above AE is parallel to BF.

$\angle ADC = 32^\circ$, $\angle BCD = 35^\circ$ and $\angle CAE = 67^\circ$.

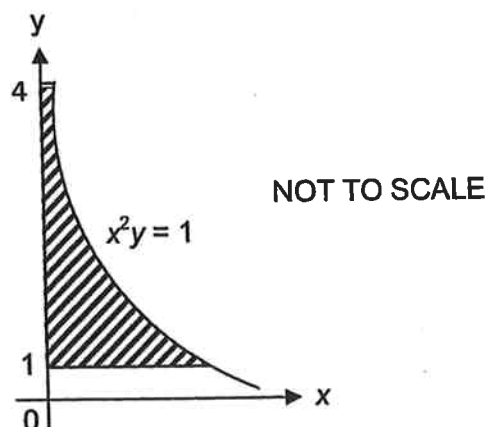
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The shaded region above shows the area bounded by the graph $x^2 y = 1$, ($x > 0$), the y -axis and the lines $y = 1$ and $y = 4$.

Find the volume of the solid of revolution formed when the shaded region is rotated about the y -axis. Give your answer in exact form.

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Question 4 continues on the next page

Question 4 (continued)

Marks

- (c) During the drought of the last few years, the water level in the local dam in the township of Wallaville was reduced to 2.5% of its capacity.

In the first week of drought breaking rain, the inflow added 3% of capacity to the amount of water in the dam (i.e. the dam was 5.5% full).

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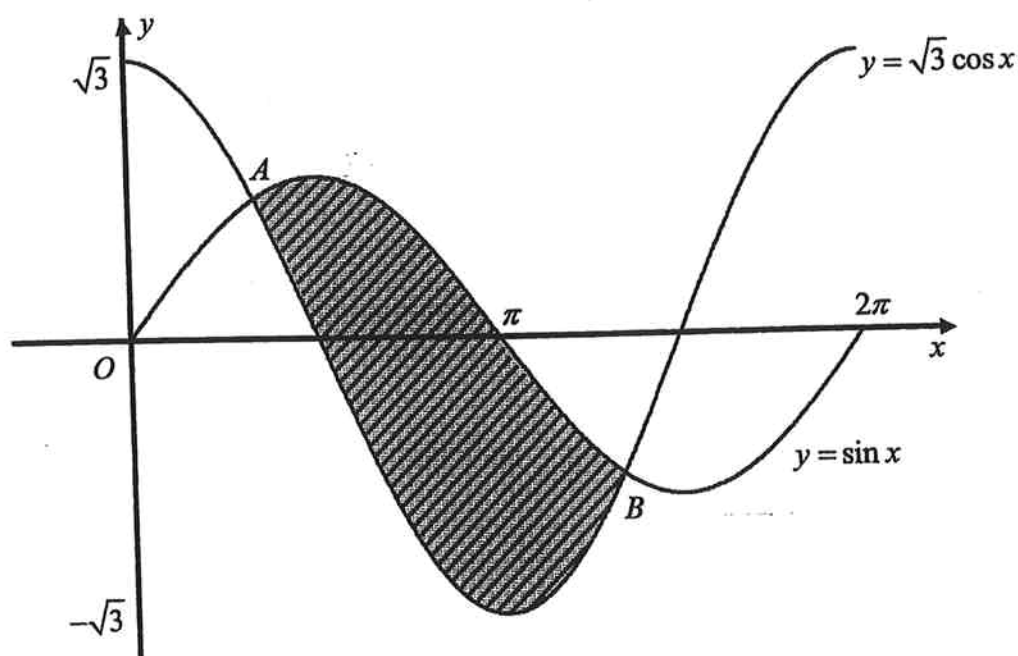
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Question 5 (12 marks) Use a SEPARATE page or writing booklet

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(a)



The diagram shows the graphs $y = \sin x$ and $y = \sqrt{3} \cos x$, $0 \leq x \leq 2\pi$.
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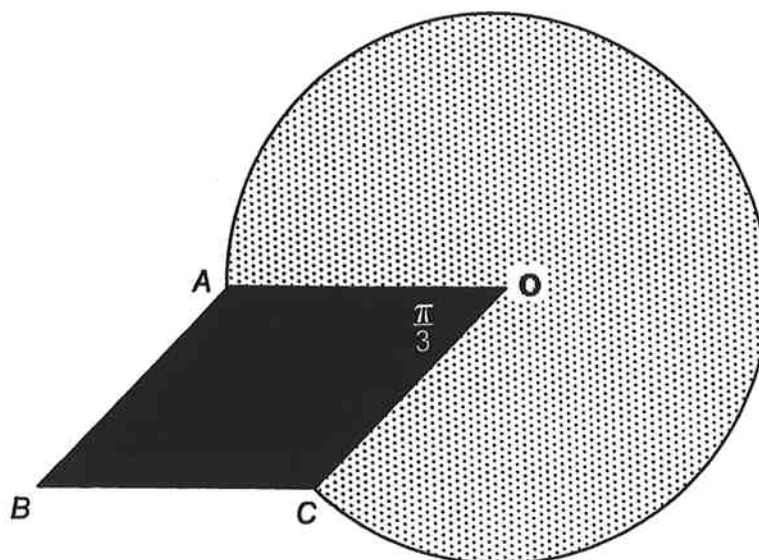
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Question 5 (continued)

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- (b) A concrete viewing platform is to be built at a mountain lookout. The platform is formed from a rhombus $AOCB$ with side $AO = 5\text{m}$ and $\angle AOC = \frac{\pi}{3}$, and the major sector of a circle centre O , radius AO . The concrete is 200mm thick. The platform is illustrated in the diagram below.

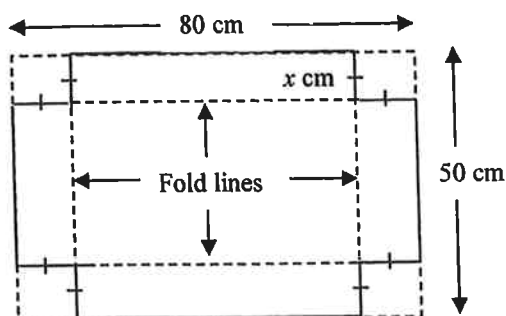


NOT TO SCALE

- (i) Show that reflex $\angle AOC = \frac{5\pi}{3}$. 1
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- (a) A piece of metal 80 cm long and 50 cm wide will be used to make an open box. A square of side x cm will be cut from each corner and the sides folded up to form a box.

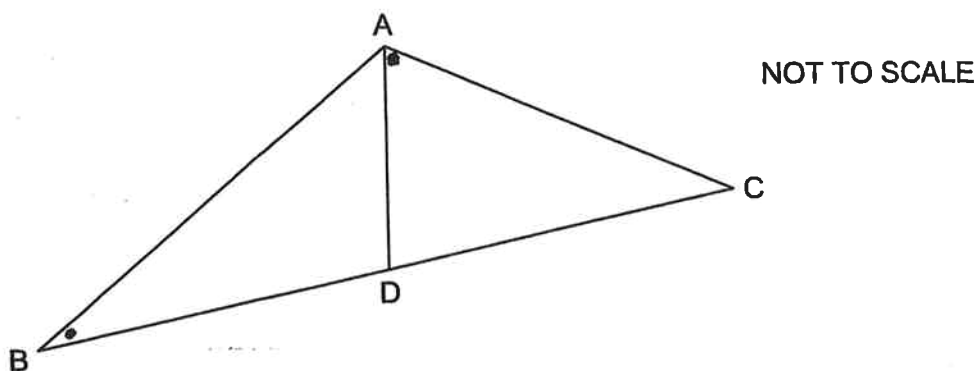


- (i) What is the limit to the maximum value of x ? Justify your answer. 1
- (ii) Show that the volume of the box is given by $V = 4000x - 260x^2 + 4x^3$. 2
- (iii) Find the value of x for which the box will have a maximum volume. 3
- (iv) Find the maximum volume of the box. 1
- (b) Evaluate: $\sum_{x=0}^4 \left(\sin \frac{\pi x}{4} \right)$ 2
- (c) Given the infinite series: $\frac{x}{3} + \frac{2x^2}{9} + \frac{4x^3}{27} + \dots$:
- (i) Show that it is a geometric series. 1
- (ii) Find the values of x such that the series has a limiting sum and find the sum (in terms of x). 2

Question 7 (12 marks) Use a SEPARATE page or writing booklet**Marks**(a) A function is defined as $f(x) = x^3 - 3x^2$.

(i) Find the coordinates of the stationary points and determine their nature. 3

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$$N = N_0 e^{kt} \text{ where } N_0 \text{ and } k \text{ are constants.}$$

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- (ii) How many worms were in the farm after 10 weeks? 1
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x	2	2.5	3	3.5	4
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- (ii) Using Simpson's Rule with 5 function values, find an approximate value for:

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- (b) The local swimming pool has been closed and is being drained for repairs. The rate, at which the amount of water in the pool, (P kilolitres) at time t hours after draining has commenced, is decreasing is given by:

$$\frac{dP}{dt} = -30(20-t)$$

Initially the pool held 6000 kilolitres.

- (i) Express P as a function of t . 2
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- (c) The equation of a parabola is: $2y = x^2 - 4x + 6$.
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- (iii) Draw a neat sketch of the graph of this parabola showing the information obtained in (i) and (ii) above. 1

Question 10 (12 marks) Use a SEPARATE page or writing booklet**Marks**

- (a) Alana has borrowed \$17 000 to buy a new car. The interest on the loan is 18% per annum paid monthly. The loan is to be repaid in equal monthly instalments of \$ P over a term of 5 years.

Let the amount owing on the loan after n months be \$ A_n .

- (i) Show that the amount \$ A_1 owing after one month is given by:

$$A_1 = \$\{(17\,000 \times 1.015) - P\}.$$

1

- (ii) Show that the amount \$ A_3 owing after 3 months is given by:

$$A_3 = \$\{(17\,000 \times 1.015^3) - P(1 + 1.015 + 1.015^2)\}.$$

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- (iii) Write down a similar expression for the amount owing after 5 years (60 months).

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- (iv) Calculate the monthly instalment \$ P paid on the loan.

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- (v) How much would Alana have saved by paying cash for the car?

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- (b) A particle is moving in a straight line so that at time t seconds its displacement from the origin is x metres. Initially the particle is 1 metre to the left of the origin.

The velocity of the particle is given by $v = 2 \cos t - 1$.

- (i) Express the displacement x as a function of t .

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- (ii) At what time is the particle first at rest?

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- (iii) Find the position of the particle at this instant.

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End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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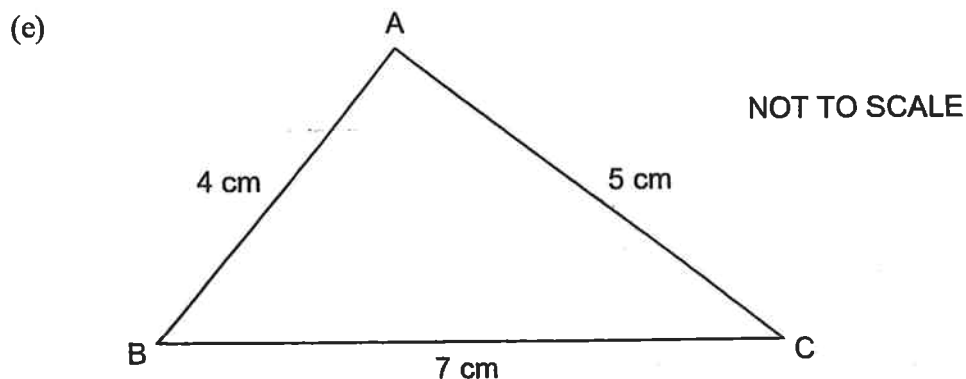
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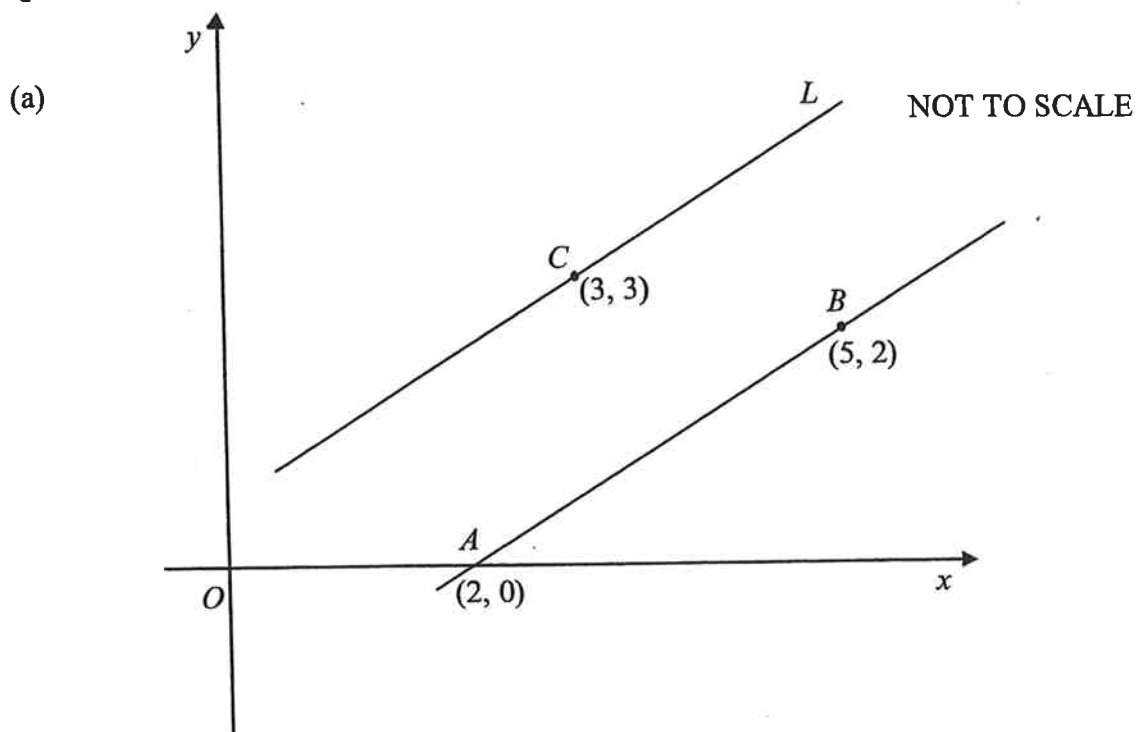
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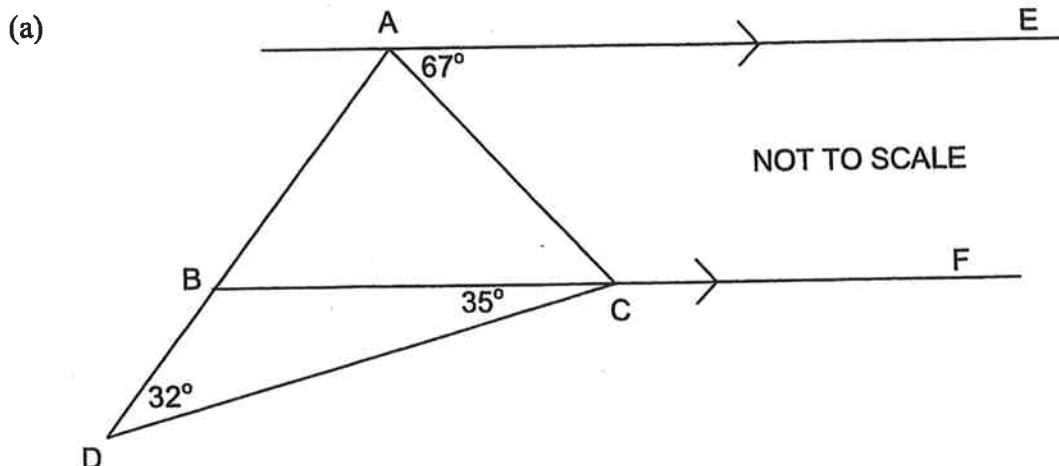
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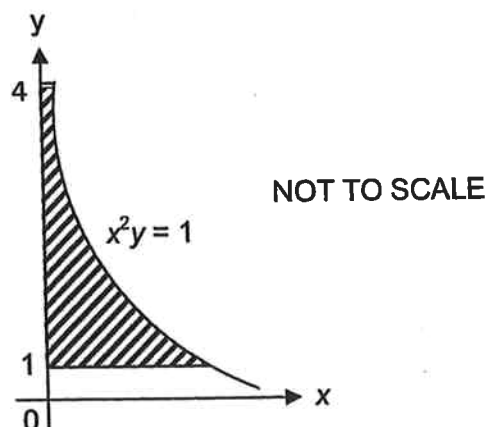
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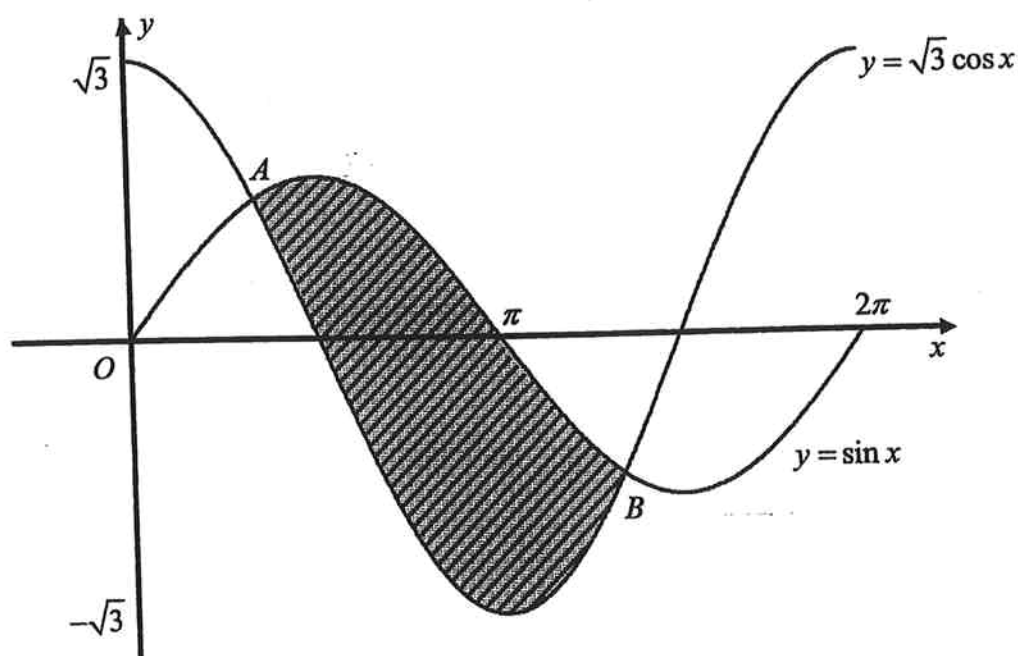
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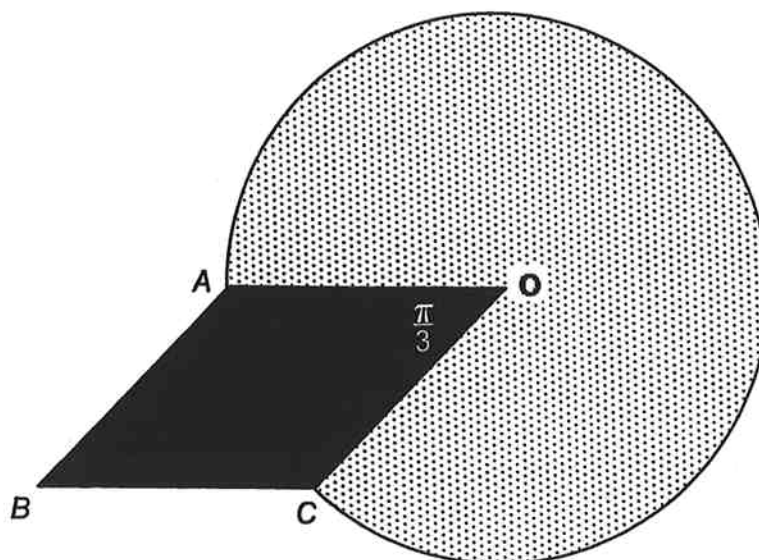
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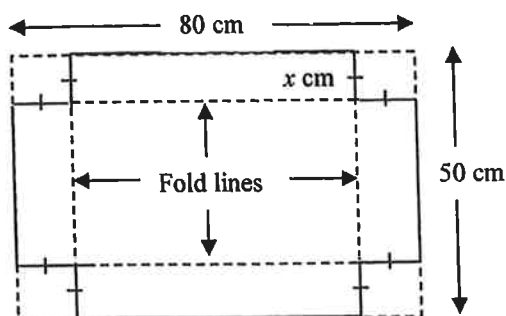


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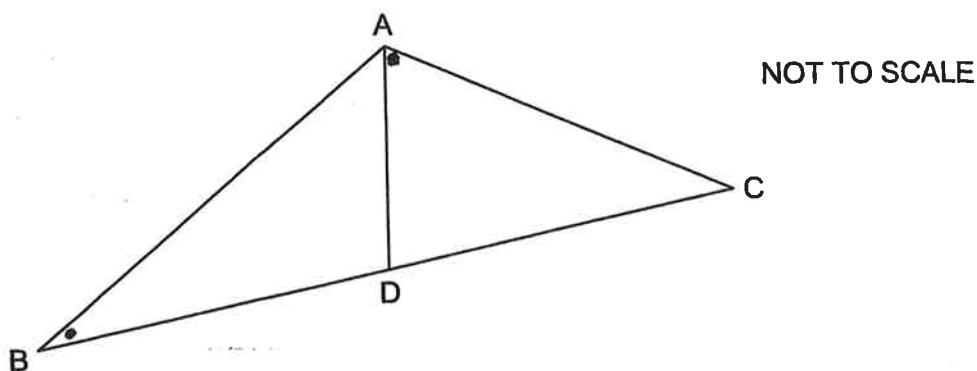


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End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$